# Lattice Diameters (of polytopes)

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## Geometric, Enumerative, & Extremal Combinatorics

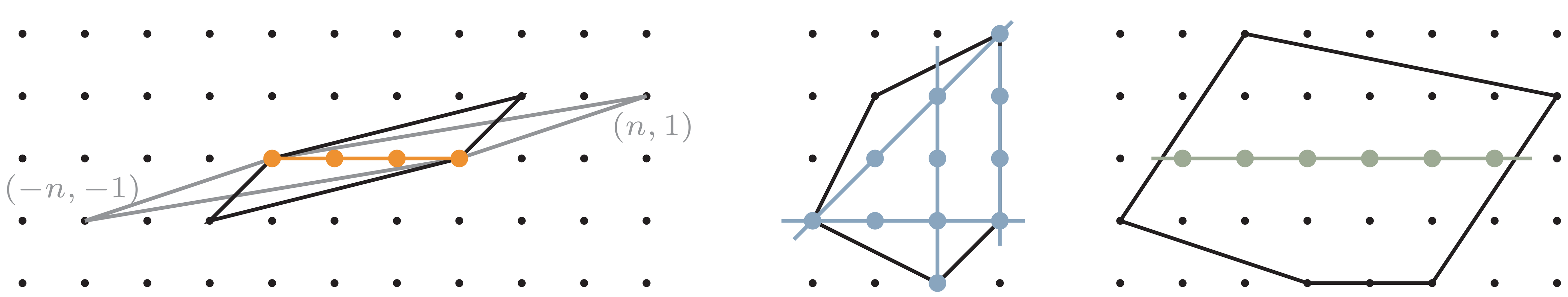
We study *lattice diameters* with respect to the following combinatorics:

- **Geometric:** lattice polytopes and rational polytopes
- **Enumerative:** Ehrhart-type theorem for subsets of lattice points
- **Extremal:** lattice-analog of *Borsuk's partition problem*

## Definition: Lattice Diameters

The **lattice diameter** of a set $S \subset \mathbb{R}^d$ is

$$\operatorname{diam}_{\mathbb{Z}}(S) := \max_{x,y \in S \cap \mathbb{Z}^d} |[x,y] \cap \mathbb{Z}^d| - 1.$$




**unrelated to Euclidean diam.** **different directions** **in interior**

## Background & Motivation

**Prior work:**

Alarcón (1987), Bárány & Füredi (2001), Averkov, Krümpelmann & Weltge (2016)

**Motivation:**

1. Lattice diameters help classify polytopes. They bound:
   - *number of lattice points* of a lattice polytope (Rabinowitz)
   - *area* of a lattice polygon
   - *lattice width* of a lattice polygon (Bárány, Füredi)
2. Related to the *shortest vector problem* and *Minkowski's successive minima.*
3. Generalizing *Ehrhart Theory* to count subsets of lattice points.
4. Study *sections* of polytopes containing maximally many lattice points.

## Theorem 1. Computation in fixed dimension

**Theorem 1.** *There exists a polynomial-time algorithm that computes a lattice diameter of a rational polytope in fixed dimension $d$.*

*Proof idea:* Solve using an integer linear programming formulation.

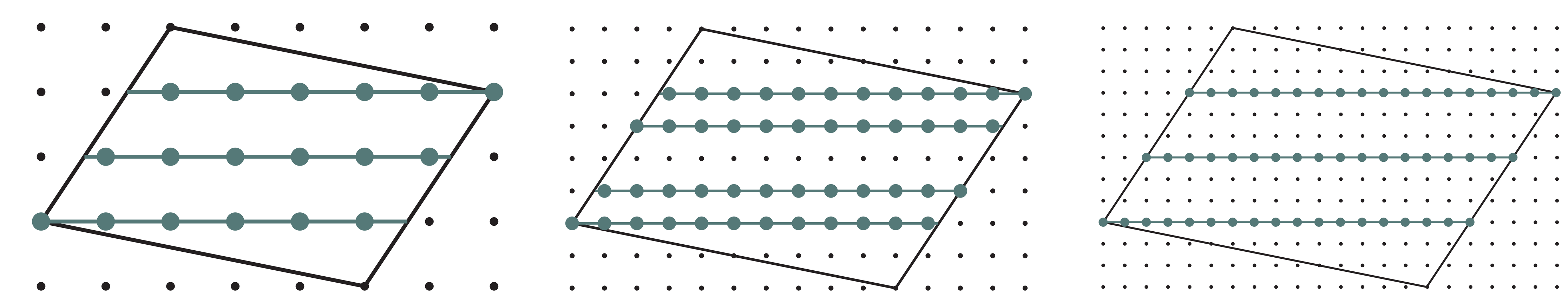
*Remark:* For *lattice* polygons (dimension two) we have a geometric algorithm.

## Theorem 2. Computation in non-fixed dimension

**Theorem 2.** *Approximating the lattice diameter of a lattice-crosspolytope in non-fixed dimension is* NP*-hard (under randomized reduction).*

*Proof idea:* Reduce from the $\ell_1$ shortest vector problem of Regev & Rosen (2006). $\operatorname{diam}_{\mathbb{Z}}(B \cdot \lozenge_d)$ yields a two-approximation of the shortest vector in $B^{-1}\mathbb{Z}^d$.

## Theorem 3. Counting Lattice Diameter Segments



$P$ $2P$ $3P$

**Def:** For $k \in \mathbb{N}$ define $\mathrm{LD}_P(k) := \#\{\text{lattice diameter segments in } kP\}$.

**Theorem 3.** *For lattice polytopes $\mathrm{LD}_P$ eventually agrees with a quasi-polynomial.*

*Remark:* In dimension two we can explicitly write out the quasi-polynomial.

## Theorem 4. Lattice Borsuk

Let $\beta_{\mathbb{Z}}(S)$ be the smallest number of parts in a partition of $S \subset \mathbb{Z}^d$ such that each part has smaller lattice diameter (discrete analogue of classical *Borsuk number*).

**Theorem 4.** $\beta_{\mathbb{Z}}(P \cap \mathbb{Z}^d) \leq 2^d$ *and this bound is best possible.*

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**References.** [1] Bárány and Füredi, *On the lattice diameter of a convex polygon*, Discrete Math., 241(1–3), 2001, Elsevier. [2] Rabinowitz, *A Theorem about collinear lattice points*, 1989. [3] Averkov, Krümpelmann and Weltge, *Notions of maximality for integral lattice-free polyhedra – the case of dimension three*, arXiv:1509.05200, 2016. [4] Regev and Rosen, *Lattice problems and norm embeddings*, Proc. 38th ACM Symp. Theory Comput. (STOC), 2006.

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